Using this coding scheme, "LET US WIN" becomes "OHW XVZ LQ". This scheme was used by Julius Caesar and is called the Caesars cipher. To decode, replace each letter by the letter three places before it. This concept is used often in Mental Ability Tests. The above can be represented as an arrow diagram as given in Figure 1.4.

This can be viewed as the set of ordered pairs

$$\{(L,O),(E,H),(T,W),(U,X),(S,V),(W,Z),(I,L),(N,Q)\}$$

which is a subset of the Cartesian product $C \times D$ where $C = \{L,E,T,U,S,W,I,N\}$ and $D = \{O,H,W,X,V,Z,L,Q\}$.



If "KDUGZRUN" means "HARDWORK", then "DFKLHYHPHQW" becomes "ACHIEVEMENT"

"Is it $f(x) = x - 3$?".

Illustration 1.2 (Geometry) Consider the following three equations

$$\text{(i) } 2x - y = 0 \quad \text{(ii) } x^2 - y = 0 \quad \text{(iii) } x - y^2 = 0$$

(i) $2x - y = 0$

The equation $2x - y = 0$ represents a straight line. Clearly the points, $(1,2),(3,6)$ lie on it whereas $(1,1),(3,5),(4,5)$ are not lying on the straight line. The analytical relation between x and y is given by $y = 2x$. The set of all points that lie on the straight line is given as $\{(x,2x) : x \in \mathbb{R}\}$. Clearly this is a subset of $\mathbb{R} \times \mathbb{R}$. (See Figure 1.5.)

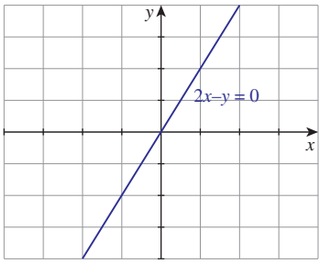


Figure 1.5

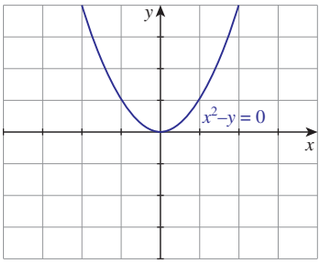


Figure 1.6

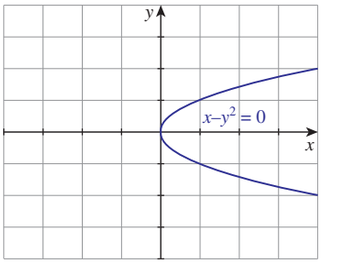


Figure 1.7

(ii) $x^2 - y = 0$.

As we discussed earlier, the relation between x and y is $y = x^2$. The set of all points on the curve is $\{(x,x^2) : x \in \mathbb{R}\}$ (See Figure 1.6). Again this is a subset of the Cartesian product $\mathbb{R} \times \mathbb{R}$.

(iii) $x - y^2 = 0$

As above, the relation between x and y is $y^2 = x$ or $y = \pm\sqrt{x},\ x \geq 0$. The equation can also be re-written as $y = +\sqrt{x}$ and $y = -\sqrt{x}$. The set of all points on the curve is the union of the sets $\{(x,\sqrt{x})\}$ and $\{(x,-\sqrt{x})\}$, where x is a non-negative real number, are the subsets of the Cartesian product $\mathbb{R} \times \mathbb{R}$. (See Figure 1.7).

From the above examples we intuitively understand what a relation is. But in mathematics, we have to give a rigorous definition for each and every technical term we are using. Now let us start defining the term "relation" mathematically.